BMJ Open

Effect of early analgesia with erector spinae plane block to reduce ventilation after severe chest trauma: the multicentre randomised controlled TARGET study protocol

Pierre Bouzat ,[1] Anais Adolle,[2] Tobias Gauss,[1] Bruno Perreira,[3] Jean-Michel Constantin[4]

To cite: Bouzat P, Adolle A, Gauss T, *et al*. Effect of early analgesia with erector spinae plane block to reduce ventilation after severe chest trauma: the multicentre randomised controlled TARGET study protocol. *BMJ Open* 2026;**16**:e117195. doi:10.1136/bmjopen-2026-117195

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (https://doi.org/10.1136/bmjopen-2026-117195).

Received 21 January 2026
Accepted 14 July 2026

Check for updates

© Author(s) (or their employer(s)) 2026. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

For numbered affiliations see end of article.

Correspondence to
Dr Pierre Bouzat;
PBouzat@chu-grenoble.fr

ABSTRACT

Introduction Blunt chest trauma is frequently associated with rib fractures, and inadequate pain control may lead to respiratory complications and prolonged ventilatory support. In spontaneously breathing patients, pain impairs coughing and secretion clearance, promoting atelectasis and respiratory failure, while in mechanically ventilated patients, it may delay weaning. Current French guidelines recommend intravenous multimodal analgesia as first-line therapy followed by thoracic epidural or paravertebral block if pain remains uncontrolled. However, these techniques have limited applicability in the trauma setting. The erector spinae plane (ESP) block is a simpler ultrasound-guided regional analgesia technique with fewer contraindications. Although retrospective data suggest potential respiratory benefits, its efficacy has not been evaluated in randomised controlled trials. This study aims to assess whether early initiation of a continuous ESP block reduces ventilatory support duration after blunt chest trauma.

Methods and analysis The EffecT of eARly analGesia with ErecTor spinae plane block (TARGET) trial is a multicentre, superiority, randomised controlled trial including adult patients (≥18 years) with blunt chest trauma and at least three rib fractures on thoracic CT scan requiring intensive care unit admission. Participants are randomised to receive either early continuous ESP block initiated within 6 hours of admission or standard intravenous multimodal analgesia with escalation to thoracic epidural or paravertebral block if indicated. The primary outcome is the number of days alive and free from invasive or non-invasive curative ventilation within 30 days after randomisation or hospital discharge. Randomisation is stratified by centre and age (<65 or ≥65 years). Investigators and outcome assessors are not blinded. A total of 400 patients will be included.

Ethics and dissemination The study has been approved by the ethics committee *CPP Sud-Ouest IV* on 17 April 2023 (approval number 23.00675.000144). The study began on 22 October 2023. Results will be disseminated through publication in a peer-reviewed international medical journal.

Trial registration ClinicalTrials.gov: NCT05920109.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This is a multicentre randomised controlled trial using a pragmatic, open-label design across 14 French intensive care units to evaluate the effect of early continuous erector spinae plane block on ventilation after blunt chest trauma.
⇒ The pragmatic study design reflects real-life trauma bay practice and enhances the external validity of the results.
⇒ The primary outcome, ventilator-free days, is a patient-centred and clinically relevant endpoint that integrates both survival and respiratory support duration.
⇒ The study is not blinded, which may introduce performance bias, although the primary outcome is objective and prospectively defined.
⇒ Crossover from the experimental group to standard analgesia may occur in case of block failure, potentially diluting the observed treatment effect.

INTRODUCTION

Blunt chest trauma is a common pattern of severe trauma patients and previous studies report a mortality rate of up to 25% following severe thoracic trauma.[1] Chest wall injuries such as rib fractures or flail chest are common lesions in this context. The diagnosis of these injuries relies on the use of thoracic CT scanning included in the whole-body CT scan. Patients with fractured ribs may require invasive or non-invasive ventilation (NIV) during their intensive care unit (ICU) stay and are at risk for several complications including pneumonia, atelectasis and acute respiratory distress syndrome.[2] Older age, greater number of fractured ribs, concomitant injuries and chronic lung disease are the main documented risk factors for complications.[3]

Pain control is one of the key goals for the management of chest trauma. Pain limits

coughing efficiency and secretion clearance, thereby potentially leading to progressive atelectasis, loss of functional residual capacity and, ultimately, respiratory distress. Pain control can improve ventilatory function and prevent respiratory complications. Diverse strategies have been proposed to alleviate pain after chest trauma. Recent French guidelines underline the benefit of regional analgesia with epidural analgesia (EA) or paravertebral block (PVB) in association with intravenous multimodal analgesia within the first 12 hours.[4] However, these techniques necessitate experienced anaesthetic medical staff and are often contraindicated in case of post-traumatic coagulopathy. As a result, only few patients benefit from these blocks in the early phase of chest trauma management and pain relief often consists of intravenous multimodal analgesia. In a recent French survey (Blondonnet *et al*[5]), for 70% of the respondents, patients with trauma with a theoretical indication of regional anaesthesia did not receive EA or PVB for the following reasons: the ICU had no experience of regional anaesthesia (62%), no anaesthesiologist-intensivist working in the ICU (46%), contraindications (27%), ignorance of the French guidelines (19%) and no protocol available (13%).[5] As a result, there is a need for a simpler regional anaesthesia that may improve pain relief and may influence outcome such as the duration of mechanical/NIV in the ICU.

Erector spinae plane (ESP) block is an alternative technique to control pain after thoracic surgery.[6] It consists in depositing the local anaesthetic in the fascial plane, deeper than the erector spinae muscle at the tip of the transverse process of the vertebra. It can be done continuously with a dedicated catheter that allows intermittent boli with a pump. It relies on a high-volume, low concentration anaesthetic, usually ropivacaine 2 mg/mL (from 20 to 40 mL). The ESP block works at the origin of the spinal nerves, based on cadaveric and contrast studies. Cadaveric studies demonstrated that block at T5–T7 level is sufficient to have unilateral sensory block from T1 to L3.[7] It can be also done bilaterally in case of bilateral injuries. This block is less invasive with fewer contraindications as compared with EA or PVBs. In particular, compared with PVB, there is no risk of pleural injury and compared with EA, it can be performed with coagulation disorders.[8]

This technique has gained interest to control postoperative pain after cardiothoracic[9] or breast surgeries.[10] After chest trauma, only a few studies have assessed the efficacy[11] and the safety[8] of ESP block. Recently, in 79 patients with traumatic rib fractures, ESP block restored respiratory capacity thanks to efficient analgesia without major complication.[12] In this study, 77% of the patients received an ESP block catheter. The catheter remained sited until the acute pain team deemed the pain could be managed orally (mean duration, 3.7 days; range 0.6–9.3 days). Incentive spirometry volumes doubled from baseline during the first 24 hours with a mean increase of 545 mL. This effect was maintained over 72 hours. Pain scores were also significantly reduced. The serratus anterior plane block (SPB) is another fascial plane technique that has been studied in rib fractures; however, its analgesic coverage is limited to the lateral chest wall (T2–T9) and it does not reliably cover posterior or medial rib fractures. In a prospective randomised trial comparing ESP block versus SPB in multiple rib fractures, ESP block provided superior analgesia and greater improvement in diaphragmatic excursion. Given this evidence and the broader anatomical coverage of the ESP block, the TARGET trial uses the ESP block as the experimental intervention. Nevertheless, there is no multicentre randomised controlled trial regarding the effect of early continuous ESP block on clinically relevant outcomes such as ventilator-free days (VFD).

We hypothesise that patients with rib fractures after chest trauma may benefit from early continuous ESP block in the trauma bay to decrease the use of invasive or NIV. The targeted population is the category of chest patients with trauma at risk of pulmonary complications: adult patients with three or more rib fractures with an indication of ICU admission.[4] We will not include patients with an extrapulmonary indication of mechanical ventilation such as patients with traumatic brain injury or spinal cord injury at the cervical or thoracic levels. We expect the early ESP block strategy to be superior to the standard strategy according to French recommendations regarding the number of VFD including non-invasive and invasive ventilation. From a public health standpoint, the use of early continuous ESP block may change current guidelines regarding chest trauma management.

This manuscript was written in accordance with the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) guidelines and the trial methodology follows recommendations from the Consolidated Standards of Reporting Trials (CONSORT) guidelines.[13]

METHODS AND ANALYSIS

The participant timeline is reported in figure 1 and in table 1 (SPIRIT figure).

Objectives

Primary objective

The main objective of this study is to assess the effect of early analgesia with continuous ESP block after chest trauma on the number of days alive and without invasive or NIV.

Secondary objectives

Comparison between the two groups of:

1. ESP block feasibility (defined as successful catheter placement, time to placement and number of attempts).
2. ESP block analgesic efficacy (pain scores, opioid consumption, spirometry parameters).
3. ESP block safety (procedure-related complications including haematoma, pneumothorax, catheter infection, haemodynamic changes).

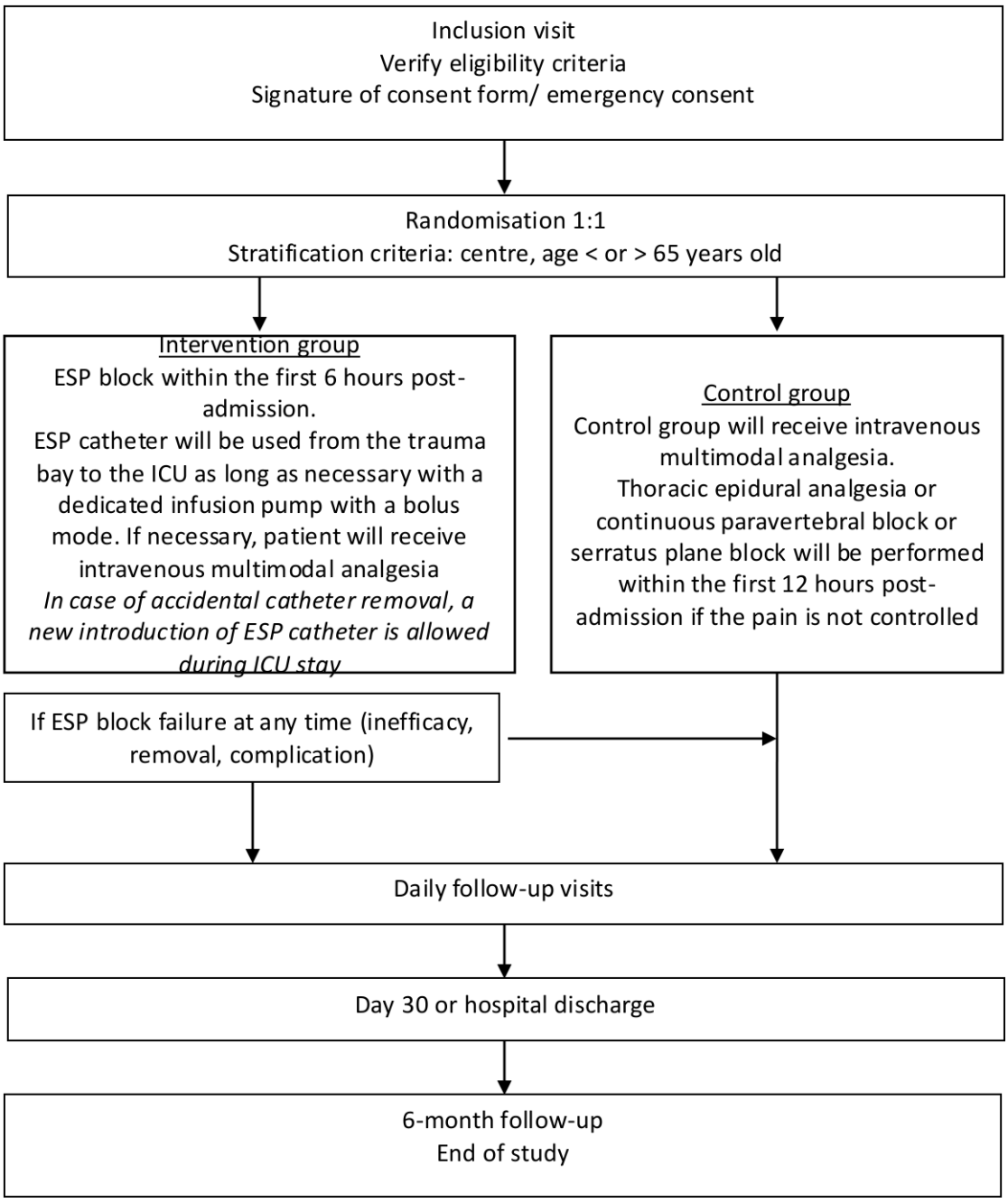


Figure 1 Participant timeline of the TARGET trial. ESP, erector spinae plane; ICU, intensive care unit; TARGET, EffecT of eARly analGesia with ErecTor spinae plane block.

4. Morbidity and mortality.

Trial design

TARGET is an open, pragmatic, multicentre, superiority randomised controlled trial and is conducted in 14 ICUs of different French hospitals. The participating centres are: CHU Grenoble Alpes (Grenoble), Hôpital Lariboisière (Paris), Hôpital de Beaujon (Paris), Hôpital La Pitié-Salpêtrière (Paris), Hôpital Européen George Pompidou (Paris), CH Annecy Genevois, CHU Bordeaux, CHU de Clermont-Ferrand, Hôpital Lyon Sud (Pierre-Bénite), Hôpital d'instruction des armées Percy (Clamart), Hôpital d'Instruction des Armées Sainte Anne (Toulon), CHU de Lille, CHU Toulouse -Rangueil, CHU Toulouse—Purpan and CHRU Tours. No financial or other incentives are offered to participants for taking part in this study.

Randomisation

Patients will be screened for enrolment as soon as possible after eligibility has been confirmed. Screening is performed by the on-call anaesthetist or intensivist present at the time of trauma bay admission, who reviews the CT scan findings and clinical status to assess inclusion and exclusion criteria. All consecutive eligible patients admitted during the study period will be screened and

Table 1 SPIRIT figure of the TARGET trial

	Trial period						
	Enrolment		Postrandomisation				Close-out
Timepoint[b]	***–24 hours* to 0 hour post-traumatic injury**	**0**	***D1***	**Daily follow-up in ICU**	***Daily follow-up in standard care service***	**Day 30 or discharge visit**	**6 months visit ± *1* month**
Enrolment:							
Eligibility screen	X						
Informed consent or emergency consent	X		X	X	X		
serum pregnancy test[†]	X						
CT scan	X						
Randomisation		X					
Intervention/comparator:							
ESP block within 6 hour posthospital admission		X	→	→	→		
Standard of care		X	X	X	X	X	
Assessments:							
Demographics, medical history		X					
Circumstances and characteristics of the injury*, thoracic AIS; ISS scores		X					
Clinical variables, numerical pain scale, spirometry parameters, haemodynamic parameters		X	X	X			
Biological parameters		X	X	X			
Ventilation status (IMV; INV), data from the locoregional analgesia		X	X	X	X	X	
SAPS II			X				
Collection of AE/SAE			X	X	X	X	X
Vital status						X	X
Quality of life questionnaire EQ-5D-5L, chronic pain assessment: VRS+DN4 general and thoracic							X

*Number of fractured ribs/flail chest/isolated chest trauma.
†For women childbearing potential*: According to CTFG guidelines, a woman is considered of childbearing potential (WOCBP), ie, fertile, following menarche and until becoming postmenopausal unless permanently sterile. Permanent sterilisation methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy. For all women with effective contraception, it is not necessary to perform βHCG testing.
AE, adverse event; AIS, Abbreviated Injury Score; CTFG, clinical trials coordination group; DN4, neuropathic pain diagnostic questionnaire; EQ-5D-L, EuroQol-5 Dimension-5 Level; ESP, erector spinae plane; HCG, human chorionic hormon; ICU, intensive care unit; IMV, invasive mechanical ventilation; INV is NIV, non invasie ventilation; ISS, Injury Severity Score; SAE, serious adverse event; SAPS, Simplified Acute Physiology Score; SPIRIT, Standard Protocol Items: Recommendations for Interventional Trials; TARGET, EffecT of eARly analGesia with ErecTor spinae plane block; VRS, verbal rating scale; WOCBP, women of childbearing potential.

a screening log is maintained at each centre to record all screened patients and reasons for non-inclusion. To promote adequate enrolment, the coordinating team at CHU Grenoble Alpes monitors inclusion rates monthly across all sites. Centres with lower-than-expected recruitment are contacted by the clinical research coordinator for case-by-case review. Regular steering committee meetings (at least every 6 months) review enrolment progress

and address any site-specific barriers to recruitment. No additional recruitment incentives (financial or otherwise) are offered to sites or participants. Enrolled patients will be randomised on a 1:1 basis in permuted blocks (variable block sizes of 2, 4 and 6) by local investigators using a dedicated, password-protected, SSL-encrypted website (CSOnline, Clinsight) accessible 24 hours a day, 7 days a week to allow immediate and concealed allocation. The centralised randomisation system automatically conceals allocation until the moment of enrolment, ensuring adequate allocation concealment. The randomisation sequence is generated and held by the data management team at CHU Clermont-Ferrand; local investigators and clinical staff are unaware of block sizes. Following enrolment in the TARGET Trial, each participant will be given a unique trial number and a case report form (CRF) will be completed by the local team. The randomisation sequence will be stratified (1) per centre and (2) according to age (<65 or ≥65 years). As the need to randomise patients during a short time-frame mandates that the randomisation is as simple as possible, we have elected not to use minimisation or to stratify on additional potential confounders. However, thoracic injury pattern (bilateral injury, flail chest, number of ribs) and ventilation status at enrolment are prespecified covariates in the adjusted analyses. The statisticians conducting the primary analysis will remain blinded to treatment allocation until the statistical analysis plan is finalised and the database is locked.

Selection of participants

Patients are included in the TARGET trial if they meet the inclusion and exclusion criteria.

Inclusion criteria

For inclusion, adult patients (aged 18 years or older) must fulfil the following criteria:

1. Blunt chest trauma with three or more rib fractures on thoracic CT scan.
2. With spontaneous breathing or under mechanical ventilation in the trauma bay.
3. Requiring an ICU admission.

Exclusion criteria

Patients fulfilling one or more of the following criteria are not included:

1. Prehospital cardiac arrest.
2. Patient not expected to survive within the first 72 hours.
3. Uncontrolled haemodynamic instability despite initial resuscitation (systolic arterial blood pressure lower than 90 mm Hg at the time of catheter insertion).
4. Extrapulmonary indication of mechanical ventilation for patients with severe traumatic brain injury (Abbreviated Injury Score, AIS, head >2).
5. Spinal cord injury at the cervical or thoracic levels.
6. Uncorrected hypovolaemia at the time of catheter insertion (defined as ongoing fluid resuscitation with vasopressor therapy or persistent haemodynamic instability not meeting the systolic arterial blood pressure threshold above).
7. Hypersensitivity to ropivacaine or other amide-bound local anaesthetics.
8. Participant currently enrolled in another interventional study within the exclusion period.
9. Pregnant, breastfeeding women.

Trial intervention

Patients in the control group will be managed according to the French recommendations for chest trauma management.[4] Patients will receive intravenous multimodal analgesia in the trauma bay. Thoracic EA or continuous PVB will be performed as soon as possible within the first 12 hours postadmission if the pain is not controlled by intravenous multimodal analgesia according to our national guidelines.

Patients in the experimental group will have a continuous ESP block within the first 6 hours postadmission. If necessary, patients will receive intravenous multimodal analgesia in the trauma bay. ESP catheter will be used from the trauma bay to the ICU as long as necessary with a dedicated infusion pump (CADD-Solis infusion pump, SMITH Medical with a bolus mode). If chest wall injuries are bilateral, bilateral ESP catheters can be done. In case of continuous ESP block failure at the early phase (defined by more than three attempts within the first 6 hours postadmission, incidence <5% of the total experimental group), patients will be switched to the control group. In case of accidental catheter removal during ICU stay, a new introduction of ESP catheter is allowed if pain is not controlled by intravenous multimodal analgesia. All protocol deviations, including crossovers and treatment switches, will be systematically documented in the eCRF with the date, reason and clinical circumstances. Crossover from experimental to control is anticipated to be infrequent (<5%), based on prior observational data; however, it will be accounted for in the per-protocol analysis. In the intention-to-treat (ITT) analysis, patients will be analysed according to their randomised allocation regardless of actual treatment received, which may dilute the observed treatment effect but preserves the integrity of the comparison. The trial is not expected to maintain complete treatment separation, as crossover is an inherent feature of pragmatic clinical trials; this is acknowledged as a limitation and will be addressed in sensitivity analyses. In the control arm, patients may receive ESP block or other regional techniques as rescue analgesia if standard therapy is insufficient, per institutional practice; this will also be documented.

ESP block procedure

After patients are positioned optimally (sitting or lateral decubitus), the affected area is identified along with the target transverse process. Given that local anaesthetic spreads cranially and caudally from the point of injection, this is usually the transverse process most central to the

affected rib levels. The ultrasound transducer is placed in a longitudinal parasagittal orientation, about 3 cm lateral to the spinous processes, allowing for visualisation of adjacent transverse processes (TP) in an in-plane approach. These are recognisable as flat, squared-off acoustic shadows with only a very faint image of the pleura visible.

After correct TP identification, an 18-gauge echogenic needle is inserted using an in-plane, cranial-to-caudal approach to contact the bony shadow of the TP with the tip deep to the fascial plane of the erector spinae muscle. The correct location of the needle tip is confirmed by injecting 0.5–1 mL of normal saline 0.9% and observing linear fluid spread lifting the erector spinae muscle off the tip of the TP. Once the fascial plane is recognised, the needle is removed and the catheter is inserted through the needle sheath. Correct catheter location is confirmed by bolusing 2–3 mL of normal saline 0.9%. Following confirmation of correct catheter tip location, 30 mL of ropivacaine 2% is injected and cranial and caudal spread of local anaesthetic can be visualised. In case of bilateral ESP catheters, initial bolus is only 25 mL by catheter. After initial bolus, a dedicated infusion pump (CADD-Solis infusion pump, SMITH Medical with a bolus mode) is used to provide a continuous infusion of ropivacaine (2 mg/mL) 1 mL/hour with a 25 mL bolus every 6 hours.

The ESP catheter is kept in place as long as it is providing analgesic benefit. Factors such as respiratory status, ambulation, oral intake of medications and chest tube removal are taken into account when deciding when to remove the ESP catheter. The catheter may also be removed if the site becomes infected, local anaesthetic leaks or at the patient's request.

ESP training

Centres have been selected because of their experience in managing patients with severe trauma. All participating centres are level I trauma centres (centres de traumatologie de niveau 1, labelled by French health authorities) with an anaesthesiologist designated as trauma team leader available 24 hours a day, 7 days a week. Each centre has confirmed a minimum annual caseload of severe trauma admissions sufficient to recruit the allocated number of participants. In case of inexperienced medical staff for this technique, the anaesthesiologist in charge of the trauma bay can be trained by a local anaesthesiologist involved in thoracic surgery. A video from Duke University is available for free to illustrate the ESP block technique: https://www.youtube.com/watch?v=O9RB0K7f8pM

Outcomes

Primary outcome

The primary endpoint is alive and VFD within the first 30 days or hospital discharge, whichever occurred first. Ventilation includes invasive and non-invasive curative ventilation but does not consist of prophylactic NIV, that is, NIV in the absence of acute respiratory failure defined as the presence and persistence for more than 30 minutes of hypoxaemia (PaO2 <60 mm Hg when breathing room air or PaO2 <80 mm Hg when breathing 15 L/min of oxygen or SpO2 <90% when breathing room air), plus either a respiratory rate higher than 30 breaths/min or clinical signs suggestive of intense respiratory muscle work and/or laboured breathing, such as use of accessory respiratory muscles, paradoxical motion of the abdomen or intercostal retraction.[14] The primary endpoint is a commonly reported composite outcome measure in acute respiratory failure trials.[15] We believe that this outcome is relevant since the tested intervention affects both the need for ventilation support and ventilator duration in the same direction as mortality. After chest trauma, VFDs appropriately capture the combined effect of continuous ESP block on both mortality and the duration of invasive/NIV given that inadequate analgesia is associated with pulmonary complications that induce morbimortality.

Secondary outcomes

For the feasibility of the intervention:

- Failure of catheter placement (more than three attempts).
- Time from admission to catheter.
- Number of attempts.

For the efficacy of the intervention:

- Numerical Rating Scale (NRS, 0–10) for pain at rest and during physiotherapy and coughing during ICU stay assessed by the bedside nurse at standardised time points. For intubated or non-verbal patients unable to self-report pain, the Behavioural Pain Scale (BPS) will be used as a surrogate measure of pain-related behaviour.
- Total opioid consumption during ICU stay, expressed in oral morphine equivalents (mg) per 24 hours and cumulated over the ICU stay.
- Spirometry parameters, collected the first 7 days of ICU stay (after extubation if mechanically ventilated).
- Chronic pain assessment at 6 months (verbal rating scale+neuropathic pain diagnostic questionnaire DN4).[16]

For the safety of the intervention:

- Haematoma after ESP block puncture.
- Pneumothorax after ESP block catheter insertion.
- Infection of the catheter during ESP block use.
- Haemodynamic parameters during ESP block use. Additional documented adverse events will include: local anaesthetic systemic toxicity, catheter dislodgement or migration, catheter occlusion, local anaesthetic leakage, unintended neuraxial spread, motor blockade and inadequate analgesia requiring rescue technique. All adverse events will be recorded by the local investigator in the eCRF and reported to the sponsor according to the regulatory framework. MedDRA coding is not mandated for this investigator-initiated trial but all serious adverse events (SAEs) will be reported as described in the safety reporting section.

For the effect of the intervention on the morbimortality:

- Number of Hospital Acquired Pneumonia during ICU stay.
- Intubation rate on day 30.
- ICU-free days within the first 30 days or hospital discharge, whichever occurred first.
- Hospital-free days within the first 30 days or hospital discharge, whichever occurred first.
- Mortality at day 1 and at day 30.
- Quality of life at 6 months (EQ-5D-5L).

Data collection

Data will be collected with a secured web-based electronic CRF (eCRF) (Ennov Clinical, Ennov, Paris, France).

The following data will be collected at the inclusion visit: demographic characteristics; comorbidities; circumstances of the accident; characteristics of the thoracic injury; Injury Severity Score (ISS) with the detailed AIS score by region; time from injury to trauma bay admission; clinical variables at the trauma bay; Glasgow Coma Scale; NRS; intubated patient; date and time if applicable; data from the locoregional analgesia (ESP block/EA/PVB (PVB)/serratus plane block (SPB): time from hospital admission to locoregional analgesia; time from ICU admission to locoregional analgesia; number of attempts/failure of catheter placement (more than three attempts); local anaesthetics and adjuvants used if applicable; complications of the locoregional analgesia; locoregional inefficacy and the reason for its inefficacy; SAEs.

The following data will be collected every day during ICU stay: Simplified Acute Physiology Score II (first day in ICU); date and time of ICU admission; NRS pain score at rest and during physiotherapy and coughing (scale 0–10); spirometry parameters: incentive spirometry during the first 7 days post ICU or after extubation if mechanically ventilated; haemodynamic parameters; clinical variables; if intubated patient: date and time of tracheal extubation(s); status of locoregional analgesia; ALR complication; morphine consumption during ICU stay; other analgesics during ICU stay (NSAIDs/ketamine/nefopam/paracetamol/tramadol); *if NIV:* reason for NIV: hypercapnia/hypoxemia/prophylactic; NIV duration.

Treatment during ICU stay (chest tube insertion: haemothorax/pneumothorax/postoperative, surgical stabilisation of ribs, any thoracic surgery); SAEs

The following data will be collected every day during hospital stay: status of locoregional analgesia; *if NIV:* reason for NIV: hypercapnia/hypoxemia/prophylactic; NIV duration

The following data will be collected at D30 or discharge: if invasive mechanical ventilation (IMV): time from hospital admission to IMV; reason for IMV; date and time of tracheal extubation(s); date and time of reintubation(s) if any; time from ICU admission to IMV; IMV duration. If NIV: reason for NIV; NIV duration. Other data collected are the SAEs, the length of stay in ICU (date and time of discharge from ICU), the length of stay in hospital (date and time of discharge from hospital), the ICU mortality, the hospital mortality

At 6 months, a phone interview will be performed by each centre to assess quality of life, chronic pain evaluation and collect adverse event.

STATISTICAL CONSIDERATIONS

Sample size

Total enrolment will target 400 patients (200 evaluable patients by randomisation group). The trial is designed with >80% power to detect a difference of 2 days alive and VFD within the first 30 days or hospital discharge whichever occurred first, with SD equals 6.5 and two-sided type I error of 5%. For this calculation, we assumed;

1. The variability of VFD would be similar to our data from our regional registry (TRENAU registry) and recent publication by Bachoumas *et al*.[2]
2. VFD does not follow Gaussian distribution.
3. A loss of follow-up negligible and a 30-day mortality around 5% according to a recent publication by Blondonnet *et al*.[5]
4. A small proportion of patients who will switch from their randomised treatment.

Statistical analysis

All analyses will be performed with the use of Stata software (V.15, StataCorp, College Station) before the breaking of the randomisation code, according to International Conference on Harmonization-Good Clinical Practice guidelines. All analyses will be detailed in a separate statistical analysis plan, which will be published before the randomisation of the last participant.

The primary analysis will be conducted in the ITT population being all randomised patients except those who withdraw consent for data use. We will perform a per-protocol analysis on the primary outcome. The criteria for including patients in the ITT and in the per-protocol populations, respectively, are provided below.

Baseline variables will be reported as numbers and percentages for categorical variables and medians with IQRs for continuous variables. According to the CONSORT 2010 statement, group differences in baseline variables will not be compared using significance testing unless specifically requested by peer reviewers.

Analysis populations

ITT population

All randomised patients except those who withdrew their consent for the use of data.

Per-protocol population

All randomised patients except patients having one or more major protocol violations defined as:

Did not receive assigned treatment,

OR

Treatment protocol mismanagement (particular attention will be paid to patients who will switch from their randomised treatment),

OR

Monitoring revealed that one or more inclusion or exclusion criteria were not met,

OR

Patients withdrawn from the protocol because the patient (or the patient's next of kin or his/her legally authorised surrogate) withdraws his/her consent.

Primary analysis

To assess the effect of early analgesia with ESP block after chest trauma on the number of days alive and without invasive or NIV, the primary analysis will be performed by non-parametric Mann-Whitney according to Yehya *et al*.[15] Non-survivors will be assigned the worst possible number of days alive and without invasive or NIV (ie, 0 days). Results will be expressed as median difference and 95% CI.

Secondary analyses

Multiple linear regression analysis will be used to identify relevant baseline covariates in the ITT population with anticipated relationship with the primary outcome (criterion for entering variables will be p<0.10), in addition to the stratification variables (age) and prespecified baseline covariates such as thoracic injury pattern (bilateral/flail chest/number of ribs), associated lesions (ISS<>15) and respiratory comorbidity. Adjusted analyses will be performed using random-effects linear regression model to take into account within and between centre variability with centre as random-effect. If appropriate, a logarithmic transformation will be envisaged to achieve normality of dependent outcome. Significance of the intervention will be assessed based on p values and regression coefficient with 95% CIs calculated from this regression.

The following prespecified post-hoc analyses will be conducted for the primary outcome (number of days alive and without invasive or NIV) for univariate and adjusted analyses:

1. The comparisons between randomisation groups will also be analysed using Fine and Gray regression with the event of interest as extubation and mortality as a competing event.[15]
2. Supplementary analysis will be conducted, in which non-survivors are omitted.
3. Given the expected non-normal distribution of VFDs and the impact of mortality, sensitivity analyses may be conducted to assess the robustness of the primary results using alternative analytical methods.

Dichotomous secondary endpoints (such as failure of catheter placement, haematoma after ESP puncture, pneumothorax after ESP catheter insertion, Hospital Acquired Pneumonia during ICU stay, infection of the catheter during ESP use, haemodynamic parameters during ESP use, intubation rate on day 30, mortality at day 1 and at day 30, numerical pain scale at rest and during physiotherapy and coughing during ICU stay categorised according to threshold reported in literature, chronic pain assessment at 6 months categorised according to threshold reported in literature) will be compared using χ^2 test and, then, random-effects robust Poisson (with log link) generalised linear model. Results will be presented as absolute differences and risk ratios with CIs. Multivariable analyses will be conducted with random-effects robust Poisson generalised linear model using the same adjustment variables described for the primary endpoint.

Continuous variables (such as time from admission to catheter, number of attempts, numerical pain scale at rest and during physiotherapy and coughing during ICU stay, opioid consumption during ICU stay, spirometry parameters, chronic pain assessment at 6 months) will be presented as mean and SDs (as median and quartiles, otherwise) and will be compared with the use of the unpaired t-test or the Mann-Whitney U test when appropriate (the Shapiro-Wilk test will be used to assess normality, and the Fisher-Snedecor test to assess homoscedasticity).

For ICU-free days and hospital-free days, generalised linear models will initially use Poisson distribution or alternatively negative binomial distribution. If assumptions for these distributions are not met, we will analyse the data using the nonparametric Van Elteren test adjusted for centre, only.

For the multivariable analyses, adjusted analyses will be performed with the use of random-effects linear or generalised linear models (for ICU-free days and hospital-free days) taking into account centre effect. When appropriate (for linear regression), a logarithmic transformation will be envisaged to achieve normality of dependent outcome. Results will be expressed as effect-sizes and 95% CIs. Adjusted analyses will be conducted using the same adjustment variables described for the primary endpoint.

Supplementary analyses of quality of life at 6 months (SF12) in each of the dimensions will be compared using a generalised linear model or a non-parametric test, with adjustment for stratification variables, and will be reported as exploratory results. Non-survivors will be assigned the worst possible scores (global and dimensions). Secondary supplementary analyses will be conducted, in which non-survivors are omitted. All exploratory results will be reported with 95% CIs.

Further details will be provided in the predefined statistical analysis plan published before the freezing of the database.

Preplanned subgroup analyses

Planned subgroup analyses will assess heterogeneity of the effect of early analgesia with ESP block after chest trauma on the primary outcome measure in prespecified subgroups of patients with thoracic injury pattern (bilateral/flail chest/number of ribs), ventilation status (non-invasive/invasive ventilation), type of surgery (thoracic/other), respiratory comorbidity (no/yes), age (<>65 years old) and associated lesions (ISS<>15).

According to usual recommendations, the interactions between group randomisation and subgroups will be evaluated in regression models (ie, random-effects linear model).

The primary outcome will be evaluated in the subgroup analyses without adjustments for multiplicity.

Significance

A two-sided p value of less than 0.05 will be considered for statistical significance of all analyses. Because of the potential for type 1 error due to multiple comparisons, findings for analyses of secondary endpoints will be interpreted as exploratory.

Interim analysis

No interim analysis will be conducted.

Method for missing data

We do not expect missing data on the primary outcome measure and a complete case analysis will be performed.

We will use an eCRF with a pragmatic design and incentive strategies to maximise complete registration in order to minimise the occurrence of missing data.

If less than 5% of the data are missing on any secondary outcomes, a complete case analysis without input of missing values will be performed. If the frequency of missing data is >5%, an additional analysis will be performed using the multiple imputation method (Stata command mi). We will use outcomes and the most important baseline characteristics in the multiple imputation. The exact variables to be used to estimate the missing values will be outlined in the detailed statistical analysis plan; if multiple imputation is used, then the primary result of the trial will be based on the imputed data.

To put significant results into perspective, a sensitivity analysis will be conducted by defining a worst-case scenario in which patients with missing data do not react to the study treatment (whatever it may be). Also, if multiple imputation is used, we will in addition provide a best–worst worst–best case scenario as a sensitivity analysis to assess the potential impact of any pattern of missing data. In the best–worst case scenario, it is assumed that patients lost to follow-up in the experimental group have had a beneficial outcome and those with missing outcomes in the control group have had a harmful outcome. Conversely, in the worst best–case scenario, it is assumed that patients who will be lost to follow-up in the experimental group have had a harmful outcome and that those lost to follow-up in the control group have had a beneficial outcome.

ETHICS AND DISSEMINATION

Research ethics approval

The TARGET study is conducted in accordance with the Declaration of Helsinki and was registered at www.clinicaltrial.gov on 26 June 2023 and last updated on 1 October 2025, under identification number NCT05920109. The trial was approved by the ethics committee *CPP Sud-Ouest IV* on 17 April 2023 (approval number 23.00675.000144, email: cppsoom4@ch-esquirol-limoges.fr) and from the National Agency for Medicines and Health Products Safety (Agence Nationale de Sécurité du Médicament et des produits de santé) on 23 March 2023. Any change to eligibility criteria, outcomes and analyses will be communicated to investigators and the ethics committee to obtain their approval.

Consent or assent

Some patients included in the protocol will be unable to consent. Written informed consent is collected from the trusted person, or the family or a close friend if they are present and able to make a free and informed decision. Otherwise, being in a life-threatening emergency situation, the investigator may waive the obligation to inform the patient until he/she deems him/her capable of giving consent (Article L1122-1-3) and proceed with the patient's inclusion. Within 24 hours, if the patient is still unable to give his consent, the trusted person (the family or a close relative) will be informed and his consent for the continuation of the study will be sought when applicable.

Thereafter, the investigating physician will have to assess the patient's aptitude to consent to the research. If the patient is able to consent, the investigating physician will be responsible for explaining the research, the allocated treatments, the risks, the foreseeable benefits and constraints, the modalities of inclusion and for collecting the patient's written informed consent. This consent strategy was approved by the ethics committee in accordance with the 2013 Declaration of Helsinki. An example of the participant information sheet and consent form is provided as online supplemental file 1.

Confidentiality

All personal data collected during this trial are pseudonymised: participants are identified only by a unique study code. No names or addresses appear in data transferred to the sponsor or data management team. Data are stored on secure, access-controlled servers at CHU Clermont-Ferrand and at the coordinating centre (CHU Grenoble Alpes). Data processing is conducted in accordance with French data protection law (Loi Informatique et Libertés, 2018) and the European General Data Protection Regulation (GDPR, EU 2016/679), within the framework of the Reference Methodology MR-001 of the Commission Nationale de l'Informatique et des Libertés (CNIL). Only authorised personnel (investigators, clinical research assistants, data managers and regulatory inspectors) have access to identifiable source data. Data will be retained for 15 years after the end of the research, in accordance with Good Clinical Practice requirements.

Blinding

This is an open-label, unblinded trial for the patient, the ICU team and the outcomes assessors.

Oversight and monitoring

Coordinating centre: Grenoble University Hospital is the sponsor of the study and is responsible for study oversight and quality controls. Steering committee: Pierre Bouzat, Tobias Gauss, Anais Adolle, Bruno Pereira, Jean-Michel Constantin. Data management team: Bruno Pereira, CHU Clermont Ferrand, France

The sponsor appointed a data safety monitoring board (DSMB) that meets every year to analyse safety data (expected and unexpected adverse events). The DSMB is composed of members independent of the research and make recommendations about protocol termination, modifications or continuation without modifications.

Adverse event reporting and harms

Adverse events are reported to the sponsor by investigating sites and are analysed by the sponsor safety departments following European regulation on clinical trials on medicinal products.

To comply with regulatory requirements regarding the reporting of serious adverse reactions to health authorities, the investigator agrees to investigate each event, meet the required notification deadlines and provide all necessary information for its analysis.

All SAEs must be reported by the investigator to the sponsor, regardless of their suspected causal relationship with the investigational strategy.

The investigator must complete and sign the SAE reporting form and attach copies of relevant laboratory test results, clinical examination reports or hospitalisation records related to the SAE. Pertinent negative results must also be included. All documents should be anonymised and must include the patient's code and study number. The initial report must be supplemented with any additional relevant information within 8 days.

All SAEs occurring in a participant enrolled in the study must be reported:

- From the date of signing the informed consent form.
- Throughout the entire follow-up period of the participant in the trial.
- After the end of the scheduled follow-up, if the event is considered possibly related to the research.

All SAEs will be reported in the trial publication.

Frequency and plans for auditing trial conduct

An audit initiated at the request of the sponsor or health authorities may be conducted at any time by trained, independent inspectors. The purpose is to ensure the quality of the research, the validity of its results and compliance with the law and regulations. The auditors/inspectors must have direct access to the source and medical data and to any useful document related to the conduct of the clinical study. The confidentiality of the data and the anonymity of the subjects will be respected.

The investigators agree to comply with the sponsor's and the competent authority's requirements for an audit or inspection of the research.

The audit may apply to all stages of the research, from the development of the protocol to the publication of results and the classification of data used or generated in the research.

Plans for communicating important protocol amendments to relevant parties (eg, trial participants, ethical committees)

Any substantial modification, that is, any modification likely to have a significant impact on the protection of participants, validity and on the results of the research, on the quality and safety of the products tested, on the interpretation of the scientific documents that support the conduct of the research or on the methods of conducting the research, is subjected to written amendment submitted to the sponsor; the latter must obtain, prior to its implementation, a favourable opinion from the ethics committee and an authorisation from the national agency for drug safety. All amendments are validated by the sponsor and by all the research stakeholders concerned by the amendment before submission to the ethic committee and the national agency for drug safety. All amendments to the protocol must be made known to all investigators participating in the research. The investigators declare compliance with the protocol. Any amendment that modifies the management of the subjects or the benefits, risks and constraints of the research is the subject of a new participant information note and a new consent form. The obtainment of informed consent follows the identical procedure as described above.

Dissemination policy

Results will be reported through scientific conferences and peer-reviewed publications following CONSORT recommendations.[17] All participants can be informed of the study results at the end of the trial on request. Any written or oral communication of research results must receive prior approval of the coordinating investigator and any committee established for the research.

The publication of the main results must indicate in the acknowledgements the name of the sponsor, and investigators, methodologists, biostatisticians and data managers who substantially participated in the research, and possibly members of the committees formed for the research, and the name of the funding source. The name of any person contributing to writing or editing the manuscript, if not already in the author list, should also be given in the acknowledgements. Publication will follow international standards of writing and publishing (The Uniform Requirements for Manuscripts of the International Committee of medical journal editors (ICMJE)) and respect the criteria for authorship.

Patient and public involvement

Patients are not involved in the design nor the conduct of the trial.

Trial status

The current protocol is version 4.0. Study started enrolment on 22 October 2023. On 6 January 2026, 386

participants are now included. Enrolment is expected to be completed by June 2026, with the final 6-month follow-up visit anticipated by December 2026.

DISCUSSION

Adequate pain management is a cornerstone of care in patients with blunt chest trauma, as uncontrolled pain contributes to respiratory complications, prolonged ventilatory support and increased morbidity. Despite guideline recommendations, the use of thoracic EA or PVB in the acute trauma setting remains limited because of technical complexity, contraindications and potential complications. The TARGET trial has been designed to address an important gap in the evidence by evaluating whether early initiation of a continuous ESP block can improve clinically relevant respiratory outcomes in this population.

The ESP block is an attractive alternative regional analgesia technique in patients with trauma. It is ultrasound-guided, relatively easy to perform, and associated with a favourable safety profile compared with neuraxial techniques. Its anatomical distance from the neuraxis and pleura reduces the risk of severe complications and allows use in patients with post-traumatic coagulopathy, a frequent limitation to epidural or paravertebral analgesia. Furthermore, the possibility of maintaining a continuous infusion via a catheter makes it suitable for prolonged analgesia from the trauma bay to the ICU. Retrospective studies and observational data have suggested potential benefits on respiratory mechanics and oxygenation, but robust prospective evidence has been lacking.

By focusing on VFDs as the primary outcome, this study integrates both survival and duration of invasive and non-invasive ventilatory support, providing a patient-centred and clinically meaningful endpoint. This outcome is particularly relevant in chest trauma, where the main therapeutic objective of analgesia is to preserve spontaneous breathing, facilitate effective coughing and secretion clearance and allow early liberation from mechanical ventilation. The pragmatic design of the trial, conducted across multiple centres and embedded in routine trauma care pathways, is expected to enhance the generalisability of the findings.

Several limitations inherent to the study design should be acknowledged. Blinding of clinicians and investigators is not feasible given the nature of the intervention, which may introduce performance bias. However, the primary outcome is objective, prospectively defined and less susceptible to subjective assessment. In addition, crossover from the experimental group to standard analgesia may occur in case of ESP block failure, potentially attenuating the observed treatment effect, although this reflects real-world practice and will be accounted for in the analysis strategy. The open-label design also introduces the risk of differential cointervention bias, whereby clinicians in the experimental group may modify analgesic or physiotherapy regimens based on their belief in the ESP block's efficacy. This is mitigated by the objective nature of the primary endpoint. Variability in cointerventions such as ventilatory strategies or physiotherapy practices between centres may also influence outcomes, although randomisation and stratification are expected to minimise imbalance. The sample size was calculated on the basis of available regional registry data (TRENAU registry) and a published observational cohort, which carry uncertainty; however, the 80% power and two-sided alpha of 5% provide a robust statistical framework. The absence of a preceding feasibility RCT is acknowledged; the decision to proceed directly to the definitive trial was based on high observed technical feasibility of the ESP block across participating trauma centres and the availability of dedicated funding. Pain assessment in non-verbal or intubated patients relies on the BPS rather than self-report, introducing some imprecision in analgesic outcome measurement. These limitations will be transparently reported in the final publication.

If early continuous ESP block is shown to reduce ventilatory support duration, this trial could support a shift in analgesic strategies for blunt chest trauma, promoting earlier use of regional analgesia in the trauma bay. Such findings would have important implications for clinical practice, resource utilisation and patient outcomes and could inform future national and international guidelines on chest trauma management.

Author affiliations
[1]Department of Anaesthesiology and Critical Care, Université Grenoble Alpe, INSERM U1216, CHU Grenoble, Alpes, France
[2]Department of Anesthesiology and Critical Care, CHU Grenoble Alpes, Grenoble, France
[3]Biostatistics Unit, Department of Clinical Research and Innovation, University Hospital of Clermont-Ferrand, Clermont-Ferrand, France
[4]Department of Anaesthesiology and Critical Care, Assistance Publique—Hôpitaux de Paris (APHP), Hôpital La Pitié Salpêtrière, Sorbonne Université, Paris, France

Contributors PB and J-MC designed the trial. BP developed the methodology and the statistical plan. TG and AA extensively revised the protocol. All authors approved the final version. PB is the guarantor.

Funding The TARGET trial was supported by funding from the French Ministry of Health (Direction Générale de l'Offre de Soins: Programme Hospitalier de Recherche Clinique National, grant number PHRC-21- 0018). The funders had no role in study design, study conduction, writing or submitting the manuscript.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval The TARGET study is conducted in accordance with the Declaration of Helsinki and was registered at www.clinicaltrial.gov on June 26 2023 and last updated on October 1 2025, under identification number NCT05920109. The trial was approved by the ethics committee CPP Sud-Ouest IV on April 17 2023 (approval number 23.00675.000144) and from the National Agency for Medicines and Health Products Safety (Agence Nationale de Sécurité du Médicament et des produits de santé) on March 23 2023. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer-reviewed.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those

of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: https://creativecommons.org/licenses/by-nc/4.0/.

ORCID iD

Pierre Bouzat https://orcid.org/0000-0003-4667-6738

REFERENCES

1 Veysi VT, Nikolaou VS, Paliobeis C, *et al*. Prevalence of chest trauma, associated injuries and mortality: a level I trauma centre experience. *Int Orthop* 2009;33:1425–33.
2 Bachoumas K, Levrat A, Le Thuaut A, *et al*. Epidural analgesia in ICU chest trauma patients with fractured ribs: retrospective study of pain control and intubation requirements. *Ann Intensive Care* 2020;10:116.
3 Huber S, Biberthaler P, Delhey P, *et al*. Predictors of poor outcomes after significant chest trauma in multiply injured patients: a retrospective analysis from the German Trauma Registry (Trauma Register DGU(R)). *Scand J Trauma Resusc Emerg Med* 2014;22:52.
4 Bouzat P, Raux M, David JS, *et al*. Chest trauma: First 48hours management. *Anaesth Crit Care Pain Med* 2017;36:135–45.
5 Blondonnet R, Begard M, Jabaudon M, *et al*. Blunt Chest Trauma and Regional Anesthesia for Analgesia of Multitrauma Patients in French Intensive Care Units: A National Survey. *Anesth Analg* 2021;133:723–30.
6 Koo C-H, Lee H-T, Na H-S, *et al*. Efficacy of Erector Spinae Plane Block for Analgesia in Thoracic Surgery: A Systematic Review and Meta-Analysis. *J Cardiothorac Vasc Anesth* 2022;36:1387–95.
7 Chin KJ, El-Boghdadly K. Mechanisms of action of the erector spinae plane (ESP) block: a narrative review. *Can J Anaesth* 2021;68:387–408.
8 White LD, Riley B, Davis K, *et al*. Safety of Continuous Erector Spinae Catheters in Chest Trauma: A Retrospective Cohort Study. *Anesth Analg* 2021;133:1296–302.
9 Marciniak D, Kelava M, Hargrave J. Fascial plane blocks in thoracic surgery: a new era or plain painful? *Curr Opin Anaesthesiol* 2020;33:1–9.
10 Singh NP, Makkar JK, Kuberan A, *et al*. Efficacy of regional anesthesia techniques for postoperative analgesia in patients undergoing major oncologic breast surgeries: a systematic review and network meta-analysis of randomized controlled trials. *Can J Anaesth* 2022;69:527–49.
11 El Malla DA, Helal RAEF, Zidan TAM, *et al*. The Effect of Erector Spinae Block versus Serratus Plane Block on Pain Scores and Diaphragmatic Excursion in Multiple Rib Fractures. A Prospective Randomized Trial. *Pain Med* 2022;23:448–55.
12 Adhikary SD, Liu WM, Fuller E, *et al*. The effect of erector spinae plane block on respiratory and analgesic outcomes in multiple rib fractures: a retrospective cohort study. *Anaesthesia* 2019;74:585–93.
13 Hopewell S, Boutron I, Chan A-W, *et al*. An update to SPIRIT and CONSORT reporting guidelines to enhance transparency in randomized trials. *Nat Med* 2022;28:1740–3.
14 Jaber S, Lescot T, Futier E, *et al*. Effect of Noninvasive Ventilation on Tracheal Reintubation Among Patients With Hypoxemic Respiratory Failure Following Abdominal Surgery: A Randomized Clinical Trial. *JAMA* 2016;315:1345–53.
15 Yehya N, Harhay MO, Curley MAQ, *et al*. Reappraisal of Ventilator-Free Days in Critical Care Research. *Am J Respir Crit Care Med* 2019;200:828–36.
16 Bouhassira D, Attal N, Alchaar H, *et al*. Comparison of pain syndromes associated with nervous or somatic lesions and development of a new neuropathic pain diagnostic questionnaire (DN4). *Pain* 2005;114:29–36.
17 Moher D, Hopewell S, Schulz KF, *et al*. CONSORT 2010 explanation and elaboration: updated guidelines for reporting parallel group randomised trials. *Int J Surg* 2012;10:28–55.